
```
file = requests.get('http://www.bradfordtuckfield.com/shakespeare.txt')
file = file.text
text = file.replace('\n', '')
print(search_suggestion('life is a', text))
```

And you can see that the suggestion is `life is a tedious`, which is the most common 4-gram that Shakespeare used that started with the words `life is a` (tied with two other 4-grams). Shakespeare used this 4-gram only once, in *Cymbeline*, when Imogen says, “I see a man’s life is a tedious one.” In *King Lear*, Edgar tells Gloucester “Thy life is a miracle” (or “Thy life’s a miracle,” depending on which text you use), so that 4-gram would also be a valid completion of our phrase.

We can have some fun by trying a different corpus and seeing how the results differ. Let’s use the corpus of Mark Twain’s collected works:

```
file = requests.get('http://www.bradfordtuckfield.com/marktwain.txt')
file = file.text
text = file.replace('\n', '')
```

With this new corpus, we can check for search suggestions again:

```
print(search_suggestion('life is a',text))
```

In this case, the completed phrase is `life is a failure`, indicating a difference between the two text corpora, and maybe also a difference between the style and attitude of Shakespeare and those of Mark Twain. You can also try other search terms. For example, `I love` is completed by `you` if we use Mark Twain’s corpus, and `thee` if we use Shakespeare’s corpus, showing a difference in style across the centuries and ocean, if not a difference in ideas. Try another corpus and some other phrases and see how your phrases get completed. If you use a corpus written in another language, you can do phrase completion for languages you don’t even speak using the exact function we just wrote.

Summary

In this chapter, we discussed algorithms that can be used to work with human language. We started with a space insertion algorithm that can